

1.1)

definice normy

$$\|u+v\|^2 - \|u-v\|^2 = \langle u+v, u+v \rangle - \langle u-v, u-v \rangle =$$

lineárta + SSS

$$= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - (\langle u, u \rangle + \langle u, -v \rangle + \langle -v, u \rangle + \langle -v, -v \rangle) =$$

lineárta + SSS ~~ještě ale~~ -1 , ~~že máme složit~~

$$= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle - \langle v, v \rangle =$$

$$= 2 (\langle u, v \rangle + \overline{\langle u, v \rangle}) = 2 \cdot 2 \operatorname{Re}(\langle u, v \rangle)$$

$$z = a + bi$$

$$\bar{z} = a - bi$$

$$z + \bar{z} = a + bi + a - bi = 2a = 2\operatorname{Re}(z)$$

$$\downarrow$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\|u-iv\|^2 - \|u+iv\|^2 = \langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle =$$

opět lineárta + SSS

$$= \langle u, u \rangle + \langle u, -iv \rangle + \langle -iv, u \rangle + \langle -iv, -iv \rangle - (\langle u, u \rangle + \langle u, iv \rangle + \langle iv, u \rangle + \langle iv, iv \rangle) =$$

$$= \cancel{\langle u, u \rangle} - \langle u, iv \rangle - \langle iv, u \rangle + \langle iv, iv \rangle - \cancel{\langle u, u \rangle} - \langle u, iv \rangle - \langle iv, u \rangle - \langle iv, iv \rangle =$$

$$= -2 (\langle u, iv \rangle + \overline{\langle u, iv \rangle}) = -2 (i \langle u, v \rangle - i \overline{\langle u, v \rangle}) =$$

$$= 2i (\overline{\langle u, v \rangle} - \langle u, v \rangle) = 2i \cdot 2i \operatorname{Im}(\langle u, v \rangle) =$$

$$= 4 \operatorname{Im}(\langle u, v \rangle)$$

$$z = a + bi$$

$$\bar{z} = a - bi$$

$$\bar{z} - z = a - bi - a - bi = -2bi = -2i \operatorname{Im}(z)$$

$$\downarrow$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$